

(a) Endpoint patents of the predicted co-citation link

US-10383765-B2

granted 2019-08-20

Apparatus and method for a global coordinate system for use in robotic surgery

co-citation degree 848

A

PREDICTED CO-CITATION

$p = 0.963$

0 co-citations observed
189 neighbours shared

US-2022101745-A1

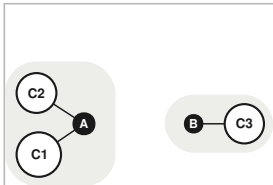
published 2022-03-31

Virtual reality system for simulating a robotic surgical environment

co-citation degree 220

B

(b) Local citation subgraph, as collected

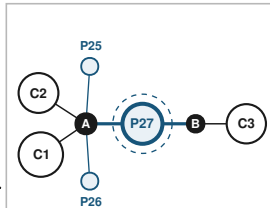


2 components

5 nodes · 3 edges

A and B sit in separate components. No seed patent cites both, so the pair is never co-cited.

(c) Prospective node inserted



1 component

8 nodes · 7 edges

P27 carries the predicted link and is the only path from A to B. The network grows to 5,265 nodes.

(d) Concept decoded from the inserted node

VEC2TEXT INVERSION · 60 STEPS · BEAM 4

Sensor-synchronized visualization of the surgical scene

One or more methods of executing a surgical system are described, including the manipulation of a virtual object by a computer-aided robotic device to generate an anatomical image of a surgical scene. The system generates an image of the surgical scene based on the user's perception of the location and orientation of the surgical scene. [...]

opening 2 of 4 sentences, verbatim

A, B = link endpoints (solid) · C1 = US-12268460-B2 (cites 610 patents) · C2 = US-11957421-B2 (246) · C3 = US-12290319-B2 (221) · P25, P26 = two further predicted links on A.

Positions: Kamada–Kawai layout of the induced subgraph, shared by (b) and (c). Node size scales with degree in the full citation network; the prospective node is drawn enlarged.

The decoded text is nearest to US-10058393-B2 (2018) within the corpus, cosine 0.951, and to US-2025288363-A1 (2025) among patents published after the collection window, cosine 0.929.